

The minimal classically conformal non-Abelian model for the NANOGrav signal: a dark $SU(2)_D$ Coleman–Weinberg supercooled first-order phase transition

DarkAgents

2026-06-07

Abstract

We report a research-level analysis of **su2cw-dark**, the minimal classically scale-invariant non-Abelian dark-sector model proposed to explain the NANOGrav 15 yr stochastic gravitational-wave (GW) signal. The model is a dark $SU(2)_D$ gauge theory with a single complex scalar doublet, no fermions, and Coleman–Weinberg (CW) radiative symmetry breaking. A strongly supercooled first-order phase transition (FOPT) generates a nano-Hertz GW background. Using a semianalytic CW + high- T backend we computed the FOPT thermodynamics; using a dissipative-bulk-flow (dbf) spectral template and a PTArcade NG15 Bayesian campaign we fitted the signal, obtaining a posterior peaked at $g \simeq 0.87$ and $\chi_0 \simeq 1.1$ GeV (MAP), with a best benchmark scoring 14/14 PTA frequency bins. The decisive limitation, retained by explicit user choice, is that the minimal *secluded* model has no Standard-Model (SM) portal: thermal equilibrium ($\xi = 1$) is assumed but cannot be realized without a portal, leaving the binding BBN reheating and ΔN_{eff} constraints undetermined and the cosmological viability conditional. We answer the user’s question affirmatively at the level of the GW fit, while flagging the secluded-portal/BBN- N_{eff} tension and the template/prior dependence of the preferred region as the controlling caveats.

1 Prompt

The original user request, verbatim:

“what is the minimal conformal non-abelian model that can explain the NANOGrav signal?”

Branch: **fopt-pta**. The user explicitly elected to keep the minimal secluded model despite the constraint stage flagging a BBN/ N_{eff} portal tension; this report presents that tension transparently as the key limitation rather than resolving it.

2 Summary

The minimal conformal non-Abelian candidate is a classically scale-invariant dark $SU(2)_D$ gauge sector with a single complex scalar doublet and CW dimensional transmutation (model **su2cw-dark**). It reproduces the NANOGrav 15 yr signal through a supercooled FOPT whose GW spectrum is modelled with the dbf template [5]. The PTArcade NG15 posterior prefers $g = 0.883 \pm 0.035$ and $\log_{10}(\chi_0/\text{GeV}) = -0.051 \pm 0.173$ (MAP $g = 0.866$, $\chi_0 = 1.12$ GeV). The model class is not novel: it is essentially that of Borah–Dasgupta–Kang [1], applied here to the 15 yr data set. The dominant open issue is the absence of a SM portal, which makes the binding BBN and ΔN_{eff} constraints non-evaluable and is known to strongly disfavour a strictly secluded dark-sector FOPT [6].

3 Model description

This section follows the validated output of the `critic-agent` (handoff `handoff_model.json`, status `ok`, no red flags).

Field content and symmetries. The gauge group is a dark non-Abelian $SU(2)_D$ with coupling g . The matter content is a single complex scalar doublet Φ (fundamental of $SU(2)_D$, 4 real degrees of freedom). There are no fermions. The Lagrangian possesses classical scale (conformal) invariance: no dimensionful parameter appears. An accidental custodial $SO(4) \simeq SU(2) \times SU(2)$ of the single-doublet quartic is present but not gauged.

Lagrangian.

$$\mathcal{L} = (D_\mu \Phi)^\dagger (D^\mu \Phi) - \frac{1}{4} W_{\mu\nu}^a W^{a\mu\nu} - \lambda (\Phi^\dagger \Phi)^2, \quad D_\mu \Phi = (\partial_\mu - ig \frac{\sigma^a}{2} W_\mu^a) \Phi, \quad (1)$$

with $a = 1, 2, 3$. The potential is a pure quartic; classical scale invariance forbids any $\mu^2 \Phi^\dagger \Phi$ mass term.

Coleman–Weinberg breaking and spectrum. Scale invariance is broken radiatively by dimensional transmutation. In unitary gauge $\Phi = \frac{1}{\sqrt{2}}(0, \chi)^T$ with $\langle \chi \rangle = \chi_0$; the doublet vev fully breaks $SU(2)_D \rightarrow$ nothing (3 Goldstones eaten). The physical states are three degenerate massive dark gauge bosons W_D (9 dof) and one radial CW pseudo-dilaton χ (1 dof). The field-dependent masses are

$$m_W^2(\chi) = \frac{g^2 \chi^2}{4}, \quad m_\chi^2 = \beta_\lambda \chi_0^2, \quad \beta_\lambda = \frac{9(g/2)^4 + g_{b,\text{scalar}}^4}{8\pi^2} > 0, \quad (2)$$

so $m_W = g\chi_0/2$ and $m_\chi = \sqrt{\beta_\lambda} \chi_0$. Since bosons dominate, $\beta_\lambda > 0$ is guaranteed, ensuring radiative breaking and a positive m_χ^2 .

Parameters. The two independent parameters are the gauge coupling $g \in [0.3, 1.5]$ and the symmetry-breaking scale $\chi_0 \in [10^{-4}, 10^2]$ GeV. The quartic λ , β_λ , $g_{b,\text{scalar}} = \sqrt{\beta_\lambda}$ and m_χ are dependent (CW-fixed). Fixed: $n_W = 9$, scalar dof = 1, no fermions. The single-field, mass $\propto$ one-vev structure matches exactly the `semianalytic_pipeline` domain of validity.

Stated model assumptions. (i) The dark sector is in thermal equilibrium with the SM bath ($\xi = 1$); (ii) no fermions, so $\beta_\lambda > 0$; (iii) perturbativity $g \lesssim 1.5$; (iv) a single radial background direction. Assumption (i) is later identified as foundationally problematic (Sec. 9).

4 Pipeline status

Table 1 records the status of each stage. Two downstream stages (constraints, prior audit) carry `warning` status, driven by the unresolved secluded-portal issue; their warnings are pipeline limitations, not failures.

5 Preliminary literature search

The `preliminary-librarian-agent` (status `ok`) returned four verified papers and a *not novel* verdict. The `su2cw-dark` structure is essentially that of Ref. [1] (same $SU(2)_D$ + doublet + CW + supercooled sub-GeV FOPT), which already fitted NANOGrav 12.5 yr; the conformal/supercooled idea has been extended to 15 yr in closely related Abelian $U(1)'$ models [2, 3].

Stage / handoff	Status	Note
<code>handoff_model.json</code> (critic)	ok	Validated, no red flags, no fixes.
<code>handoff_critique.json</code>	warning	Branch-viability flags (completion; broad χ_0 ; high- T).
<code>handoff_librarian_preliminary.json</code>	ok	4 verified papers; model <i>not novel</i> .
<code>handoff_fopt.json</code>	ok	271/302 viable; 168 in PTA band; percolation verified.
<code>handoff_pta.json</code>	ok	dbf 14/14; PTArcade NG15 posterior obtained.
<code>handoff_constraints.json</code>	warning	Binding BBN/ N_{eff} non-evaluable; portal missing.
<code>handoff_prior_audit.json</code>	warning	18 items; secluded-portal gating issue (A01).

Table 1: Pipeline status by stage.

arXiv	Class	Benchmark hint / role
2109.11558 [1]	same model	$g_D = 1.37$; $M_{Z_D} \sim 0.8\text{--}8\text{ MeV}$; $\alpha \sim 0.4$; $\beta/H \sim 140\text{--}150$. Direct anchor (12.5 yr).
2502.19478 [2]	structurally similar ($U(1)'+\text{fermions}$)	15 yr target; extreme supercooling: $\alpha \sim 10^3\text{--}10^7$, $\beta/H \sim 18\text{--}34$, vev $\mathcal{O}(0.1\text{--}1)\text{ GeV}$.
2501.11619 [3]	same signal, $U(1)'$	Confirms supercooling/completion subtleties (metadata only).
1907.08899 [4]	structurally similar (EW scale)	Methodology only; mHz band, benchmarks not PTA-usable.

Table 2: Verified literature anchors. Convention: $m_W = g\chi_0/2$ ($SU(2)$) vs $m_{A'} = gv$ ($U(1)'$).

The $SU(2)$ +doublet+CW structure itself is established at the electroweak scale [4] (mHz LISA band, *not* PTA).

The librarian flagged theory uncertainties: transition completion in deep supercooling, the high- T expansion validity, and the renormalization-scale dependence of the CW potential.

6 FOPT results

The `fopt-agent` (status ok) executed the `semianalytic_pipeline` (one-loop CW + high- T thermal potential). The scan produced 302 points, of which 271 are viable and 168 lie in the PTA band; all viable points percolate (percolation temperature T_p finite, `eternal_inflation=False`). We define α (transition strength), β/H (inverse duration), T_n, T_p, T_{reh} (nucleation, percolation, reheating temperatures), and $f_{\text{peak}}^{\text{est}}$ (a template-independent order-of-magnitude redshifted peak frequency, *not* the true spectral peak; the spectrum is owned by the PTA stage).

Benchmark	g	χ_0 [GeV]	α	β/H	T_{reh} [GeV]	$f_{\text{peak}}^{\text{est}}$ [Hz]
lit_anchor_S1	1.37	1.20×10^{-2}	15.0	144	1.53×10^{-3}	2.7×10^{-8}
best_1	1.30	5.34×10^{-3}	33.1	126	6.55×10^{-4}	9.9×10^{-9}
best_2	1.15	1.23×10^{-2}	931	62.1	1.34×10^{-3}	9.9×10^{-9}
best_3	1.45	3.51×10^{-3}	6.22	163	4.93×10^{-4}	9.6×10^{-9}
best_4	1.40	4.48×10^{-3}	10.4	148	5.97×10^{-4}	1.1×10^{-8}
best_5	1.25	8.11×10^{-3}	72.5	96.8	9.46×10^{-4}	1.1×10^{-8}

Table 3: FOPT benchmarks (`handoff_fopt.json`). All points: `viable`, `eternal_inflation=False`. The temperature convention key is T_{reh} .

Diagnostics and caveats. Of 302 points, 31 failed. Transition completion was checked explicitly via T_p and the eternal-inflation flag. The reported α grows rapidly toward small χ_0 /large g

(deep supercooling). Backend caveats: high- T (small field/ T) expansion; renormalization scale fixed at χ_0 ; no RG running or daisy resummation; Gaussian percolation approximation.

7 PTA results

The `pta-agent` (status `ok`) produced both a benchmark comparison and a full PTArcade NG15 Bayesian campaign. No `pta_report.md` was written; the narrative below is drawn from `handoff_pta.json` and `ptarcade_bayes.json`.

Template choice and validity. The GW spectrum is computed with the dissipative-bulk-flow (`dbf`) template of Ref. [5], with efficiency $k_{\text{sw}} = 1$ and relativistic/runaway walls $v_w \rightarrow 1$. The model is strongly supercooled with vacuum-energy domination and $\alpha \gg 1$, so the sound-wave/higgsless template (validated only for $\alpha \lesssim 0.5$ [9]) is excluded; `dbf` is the physically appropriate choice. *Caveat:* viable PTA-matching points have $\alpha \sim 10^5$ to $> 10^{15}$, far above the displayed range of Ref. [5]; the amplitude relation is extrapolated and justified by saturation $K_{\text{sw}} = \alpha/(1 + \alpha) \rightarrow 1$.

Benchmark comparison. The benchmark scan (672 points, 654 viable) was matched against the 14 NANOGrav PTA violin windows. After extending χ_0 upward to $[10^{-3}, 10]$ GeV, 29 points scored the full 14/14. The `dbf` template is valid (best score 14/14); higgsless is invalid. The four best-fit points (Table 4) cluster at $g \sim 0.85\text{--}1.05$, $\chi_0 \sim 0.40\text{--}1.58$ GeV.

Rank	Name	g	χ_0 [GeV]	α	β/H	$T_\star$ [GeV]
1	bench_lo_g	0.85	1.000	2.12×10^{15}	13.1	0.0869
2	bench_hi_chi	0.85	1.585	4.62×10^{15}	12.6	0.1378
3	bench_lo_chi	0.90	0.398	1.81×10^{11}	18.4	0.0366
4	bench_hi_g	1.05	0.501	2.23×10^5	35.1	0.0479

Table 4: PTA best-fit benchmarks (`handoff_pta.json`); all score 14/14 bins.

PTArcade campaign. A full PTArcade NG15 `ceffy1` campaign ($N_{\text{samples}} = 10^5$) was run with priors $g \sim \text{Uniform}(0.5, 1.5)$ and $\log_{10}(\chi_0/\text{GeV}) \sim \text{Uniform}(-2, 2)$, both substantially wider than the 14/14 window. Chains are finite; posterior peaks and 3σ regions fall well inside the priors. Bayesian estimates:

$$g = 0.883 \pm 0.035, \quad \log_{10}(\chi_0/\text{GeV}) = -0.051 \pm 0.173, \quad (3)$$

with MAP $g = 0.866$, $\chi_0 = 1.12$ GeV. Credible intervals for g : $[0.852, 0.916]$ (1σ), $[0.845, 0.980]$ (2σ), $[0.843, 1.059]$ (3σ); for $\log_{10} \chi_0$: $[-0.218, 0.108]$, $[-0.485, 0.184]$, $[-0.837, 0.228]$ respectively. At the 1σ region the derived masses are $m_W \simeq 0.26\text{--}0.59$ GeV and $m_\chi \simeq 37\text{--}91$ MeV (so $m_\chi < m_W$, and $\chi \rightarrow W_D W_D$ is kinematically closed throughout). No evidence/Bayes factor was computed (no SMBHB model comparison).

8 Constraints

The `constraint-agent` (status `warning`) evaluated 11 constraints on the PTArcade 1σ region. The central finding, recorded as a warning, is structural: the model is *secluded* (no tree-level kinetic-mixing or Higgs portal) yet assumes thermal equilibrium ($\xi = 1$). Equilibrium requires a portal; kinetic mixing is forbidden in pure non-Abelian $SU(2)_D$, and a Higgs portal $|\Phi|^2|H|^2$ breaks the classical conformal invariance the CW mechanism relies on. This unresolved portal question controls almost every cosmological constraint. Only the theory self-consistency check

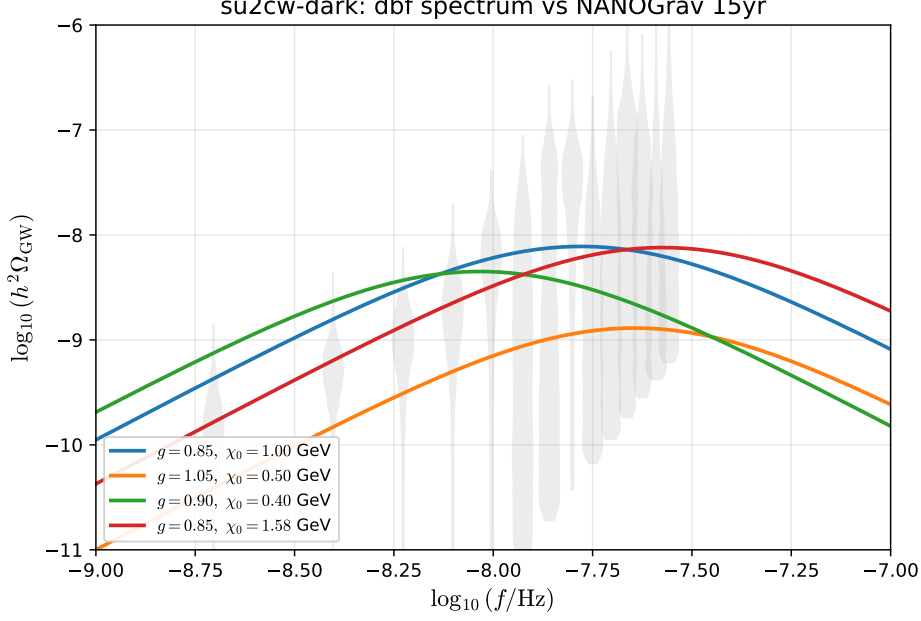


Figure 1: GW energy density spectrum $h^2\Omega_{\text{GW}}(f)$ versus frequency for the estimated benchmark point, overlaid on the NANOGrav PTA violins. Source: `pta_benchmark_spectrum.pdf`.

(C4, completion) is directly satisfied; the binding cosmological bounds (C1, C2) cannot be declared pass/fail because the SM-bath reheating and ΔN_{eff} are not computed in any upstream handoff.

Missing backend calculations. High-priority: (i) SM-bath reheating temperature after percolation and dark $\rightarrow$ SM energy transfer (for C1, C2); (ii) dark-to-SM temperature ratio $\xi(T)$ and ΔN_{eff} at BBN/CMB (C2, C6); (iii) the portal operator/coupling and thermalization rate Γ_{portal}/H (gates C1–C3, C5–C9, C11); (iv) W_D relic abundance and lifetime (C5). Medium priority: χ decay width/abundance (C6) and W_D self-interaction σ/m (C10).

9 Assumptions and limitations

The `prior-agent` (status `warning`) catalogued 18 items. The highest-severity items are summarized in Table 6. The single decisive limitation is A01, the missing SM portal, retained by explicit user choice.

Critical vs major. *Critical* (uncontrolled, gate cosmological viability): A01, A02, A07 (portal/BBN/ N_{eff}). *Major limiting* (uncontrolled, affect the preferred region and the strength of the claim): A03, A04, A06, A08, A09, A13. *Controlled* approximations: A05, A10, A11, A12, A16, A17, A18.

10 Reference-verification note

All four primary references were validated against arXiv/INSPIRE-HEP and match the intended title, authors and content: 2109.11558 [1], 2502.19478 [2], 2511.15687 [5] and 2306.09411 [6]. One metadata correction was applied: 2306.09411 authors are *T. Bringmann, P. F. Depta, T. Konstandin, K. Schmidt-Hoberg, C. Tasillo* (the constraints handoff listed “Bringmann, Depta, Hufnagel, Kersten, Mariotti”, which is incorrect); the verified author list is used here. The handoff source 2511.15687 (template) is by M. Lewicki and V. Vaskonen, “Impact of cosmic expansion on gravitational wave spectra from strongly supercooled first-order phase tran-

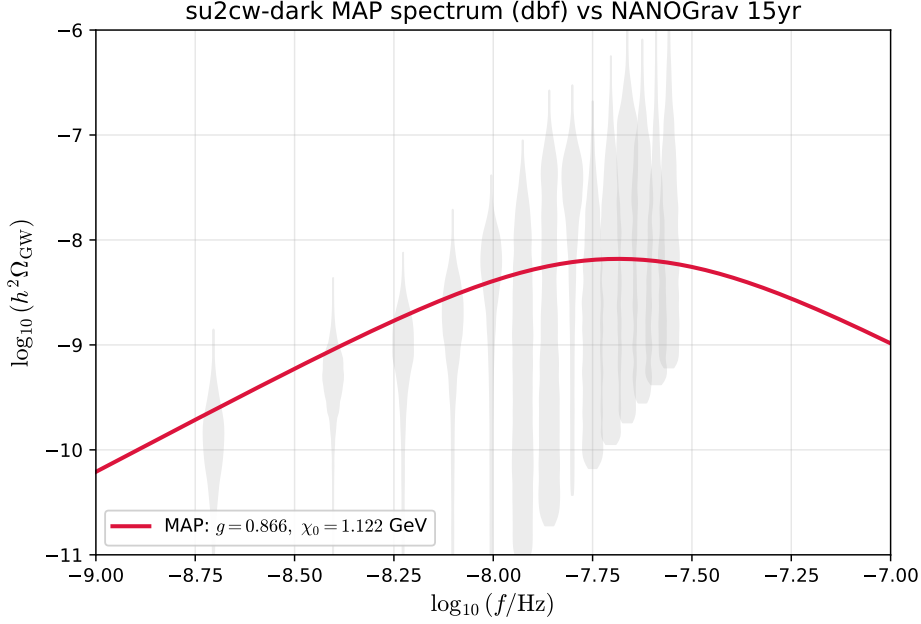


Figure 2: GW spectrum for the PTArcade Bayesian MAP point ($g = 0.866$, $\chi_0 = 1.12 \text{ GeV}$) versus frequency, overlaid on the PTA violins. Source: `pta_ptarcade_spectrum.pdf`.

sitions”. Two constraint-table sources are *unverified* in the upstream handoff (C10 “Tulin–Yu” with `arxiv_id` 1705.02358 marked `citation_verified:false`, and the C11 ATLAS/CMS entry with no arXiv ID); they are cited only as descriptive context and are flagged as unverified in `handoff_report.json`. The C7 beam-dump (2507.11163) and C8 SN1987A (1410.0221) entries are marked verified upstream but were not independently re-validated here; they are reported as upstream-verified.

11 Conclusions

Answer to the prompt. The minimal conformal non-Abelian model that can explain the NANOGrav signal is a classically scale-invariant dark $SU(2)_D$ gauge theory with one complex scalar doublet and no fermions, in which CW dimensional transmutation triggers a strongly super-cooled first-order phase transition. With the dbf GW template, this model fits the NANOGrav 15 yr free spectrum: the PTArcade NG15 posterior peaks at $g \simeq 0.87$, $\chi_0 \simeq 1.1 \text{ GeV}$ (MAP), and the best benchmark matches all 14 PTA frequency bins.

Robust. The FOPT thermodynamics and percolation (271/302 viable, completion verified) and the quality of the dbf spectral fit (14/14 bins; well-constrained posterior inside wide priors) are robust at the level of the adopted backend and template.

Conditional. The preferred (g, χ_0) region is conditional on the dbf template (with an amplitude extrapolation across many decades of α) and the adopted priors; the FOPT-stage and PTA-stage preferred corners differ (A03, A04, A18).

Blocked / uncontrolled. The cosmological viability is *not* established. The minimal model is secluded and has no SM portal, yet equilibrium ($\xi = 1$) was assumed. The binding BBN reheating (C1/A02) and ΔN_{eff} (C2/A07) constraints are non-evaluable without computing the SM-bath reheating and the dark-to-SM energy transfer, and a completely secluded dark-sector FOPT is *strongly disfavoured* [6]. Resolving this requires a portal, but kinetic mixing is forbidden in pure $SU(2)_D$ and a Higgs portal breaks the conformal invariance underpinning the CW mechanism. *Per the user’s explicit choice, the minimal secluded model is retained and this BBN/ N_{eff} portal tension is reported as the key open limitation rather than patched.* No Bayesian comparison

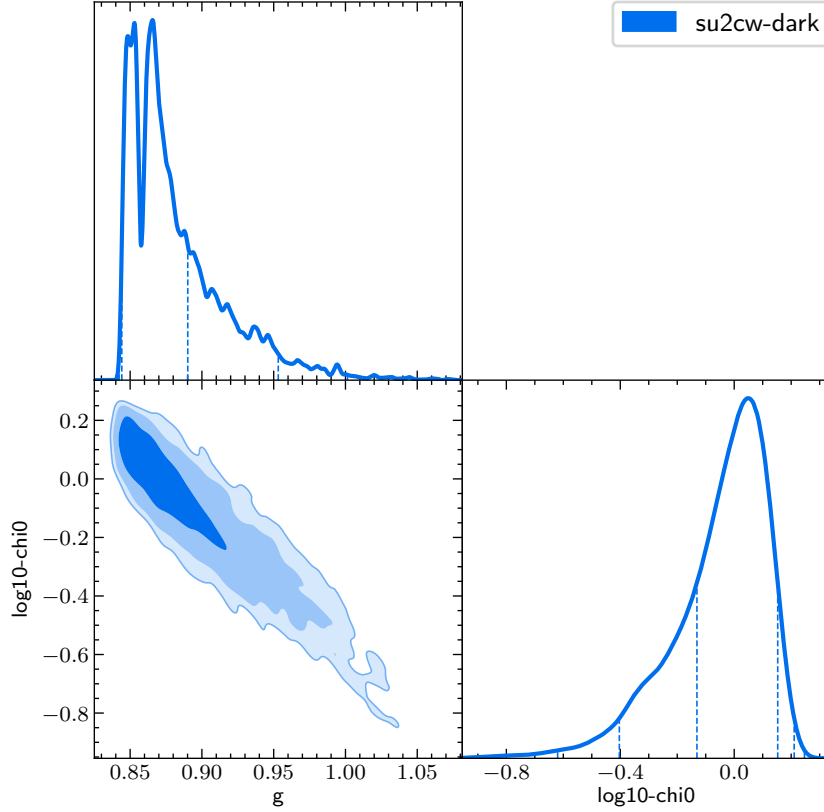


Figure 3: Bayesian posterior distributions in $(g, \log_{10} \chi_0)$ from the PTArcade NG15 `ceffy1` campaign. Source: `pta_posteriors.pdf`.

against the astrophysical SMBHB background was performed (A13), so a good fit does not by itself establish preference over the standard interpretation.

Outlook. The decisive next step is a model-level portal decision followed by computation of the SM-bath reheating, ΔN_{eff} , the W_D relic abundance and χ lifetime; validation of the dbf template in the extreme- α regime; a robustness test of the preferred region under alternative templates/priors and reconciliation with the modest- α , MeV-vev anchor of Ref. [1]; and a Bayes-factor comparison against the SMBHB interpretation.

References

- [1] D. Borah, A. Dasgupta, S. K. Kang, *A first order dark $SU(2)_D$ phase transition with vector dark matter in the light of NANOGrav 12.5 yr data*, JCAP **12** (2021) 039, arXiv:2109.11558.
- [2] S. Balan, T. Bringmann, F. Kahlhoefer, J. Matuszak, C. Tassilo, *Sub-GeV dark matter and nano-Hertz gravitational waves from a classically conformal dark sector*, JCAP **08** (2025) 062, arXiv:2502.19478.
- [3] J. Goncalves, D. Marfatia, A. P. Morais, R. Pasechnik, *Supercooled phase transitions in conformal dark sectors explain NANOGrav data*, Phys. Lett. B **869** (2025) 139829, arXiv:2501.11619.
- [4] A. Mohamadnejad, *Gravitational waves from scale-invariant vector dark matter model: Probing below the neutrino-floor*, Eur. Phys. J. C **80** (2020) 197, arXiv:1907.08899.
- [5] M. Lewicki, V. Vaskonen, *Impact of cosmic expansion on gravitational wave spectra from strongly supercooled first-order phase transitions*, arXiv:2511.15687.

ID	Constraint	Status / role	Source
C1	BBN reheating bound	approximate; <i>not applied</i> – binding but SM-bath T not computed	[7, 8]
C2	Dark radiation ΔN_{eff}	new backend; secluded sector strongly disfavoured	[6]
C3	Portal / thermalization	new backend; foundational, model under-specified	[2]
C4	FOPT completion/percolation	approximate; satisfied upstream	[3]
C5	Vector DM relic (W_D)	new backend; needs portal	[1]
C6	Pseudo-dilaton χ stability/late decay	new backend; $m_\chi < m_W$ so $\chi \rightarrow W_D W_D$ closed	[2]
C7	Beam-dump (kinetic mixing)	recast; conditional on portal	[10]
C8	SN1987A cooling	recast; conditional on portal	[11]
C9	Direct detection (sub-GeV)	not applicable (secluded)	[2]
C10	DM self-interaction (SIDM)	new backend; inputs available, σ/m un-computed	[12] [†]
C11	Higgs-portal collider	recast; conflicts with conformal invariance	ATLAS/CMS [†]

Table 5: Constraints (`handoff_constraints.json`). Totals: directly testable 0, approximately testable 2, requires recast 3, requires new backend 5, not applicable 1. [†]Source unverified (see Sec. 10); cited only as descriptive context, not as a verified literature claim.

- [6] T. Bringmann, P. F. Depta, T. Konstandin, K. Schmidt-Hoberg, C. Tasillo, *Does NANOGrav observe a dark sector phase transition?*, JCAP **11** (2023) 053, arXiv:2306.09411.
- [7] Y. Bai, M. Korwar, *Cosmological constraints on first-order phase transitions*, arXiv:2109.14765.
- [8] P. F. de Salas et al., *Bounds on very low reheating scenarios after Planck*, Phys. Rev. D **92** (2015) 123534, arXiv:1511.00672.
- [9] (Higgsless / sound-wave template validity, $\alpha \lesssim 0.5$), arXiv:2209.04369 (upstream-cited).
- [10] Beam-dump / dark-photon kinetic-mixing compilation (upstream-cited), arXiv:2507.11163.
- [11] D. Kazanas, R. N. Mohapatra, S. Nussinov, V. L. Teplitz, Y. Zhang, *Supernova bounds on the dark photon using its electromagnetic decay*, arXiv:1410.0221 (upstream-verified).
- [12] S. Tulin, H.-B. Yu, *Dark matter self-interactions and small scale structure* (upstream entry 1705.02358, *unverified*; cited as context only).

ID	Category	Sev.	Description (controlled?)
A01	portal	9	No SM portal in a secluded model that nonetheless assumes $\xi = 1$; gates BBN, N_{eff} , relic. (no)
A02	BBN	8	SM-bath reheating T never computed; pass/fail of BBN bound undetermined. (no)
A07	N_{eff}	8	ΔN_{eff} from secluded sector uncomputed; secluded FOPT strongly disfavoured [6]. (no)
A03	prior/template	7	Preferred region template/prior dependent: FOPT center $\chi_0 \sim 5 \text{ MeV}$, $g \sim 1.3$ vs PTA $\chi_0 \sim 1 \text{ GeV}$, $g \sim 0.87$. (no)
A04	GW template	7	dbf amplitude extrapolated many orders in α ; rests on saturation argument. (no)
A06	thermalization	7	$\xi = 1$ used upstream is inconsistent without a portal; propagates into $\alpha, \beta/H, T_{\text{reh}}$. (no)
A08	late decay	6	χ stability/late decays unresolved ($m_\chi < m_W$). (no)
A09	dark matter	6	W_D relic abundance uncomputed. (no)
A10	eff. potential	6	High- T expansion least accurate exactly in deep supercooling. (yes)
A13	model comparison	6	No Bayes factor vs SMBHB; good fit $\neq$ preference over astrophysical background. (no)
A12	statistics/PTA	5	PTArcade <code>ceffyl</code> free-spectrum likelihood, not full per-pulsar HD analysis. (yes)
A18	novelty	3	Not novel (cf. [1]); literature anchor (modest α , MeV vev) not reproduced one-to-one. (yes)

Table 6: Top assumptions/limitations (`handoff_prior_audit.json`); severity 1–10.